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Political Uncertainty and Household Stock Market Participation

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Research questions

- ◆ Does political uncertainty (PU) affect the propensity and intensity of households' SMP?
- ◆ If so, where do households reallocate their capital?
- ◆ What are the potential channels through which political uncertainty affects SMP?
- ◆ Does SMP reverse after PU resolves? If so, does it reverse completely or partially?



What does PU seek to capture?

- ◆ Uncertainty about *who* will make policy decisions (e.g., who will win the next elections)
- ◆ Uncertainty about *what* economic policy actions the decision makers will undertake, and *when*
- ◆ Uncertainty about the economic *effects* of policy actions (consequences of past, present, and future actions)
- ◆ Uncertainty induced by policy *inaction* (policy delays)



Measuring PU – using gubernatorial elections

- ◆ State governments have substantial power in shaping the state's economic environment through policies on taxes, subsidies, state budget, and wages
- ◆ Pre-scheduled and not controlled or affected by households (exogenous events) => mitigates endogeneity concerns that changes in equity participation may be caused by changes in business cycles or in state economic conditions
- ◆ Used in various studies including Atanassov, Julio, and Leng, 2016; Bird, Karolyi, and Ruchti, 2017; Çolak, Durnev, and Qian, 2017; Jens, 2017



Gubernatorial vs. Presidential Elections

- ◆ While presidential elections create nationwide political uncertainty, gubernatorial elections lead to statewide political uncertainty that have stronger localized effects
- ◆ Unlike presidential elections that are held once every four years nationwide, gubernatorial elections in different states are staggered and held in different years => enables us to implement a DD approach which helps address omitted variables problems
- ◆ State governors have varying term lengths and term limits => Create important cross-sectional and time-series variations that can help better identify the effect

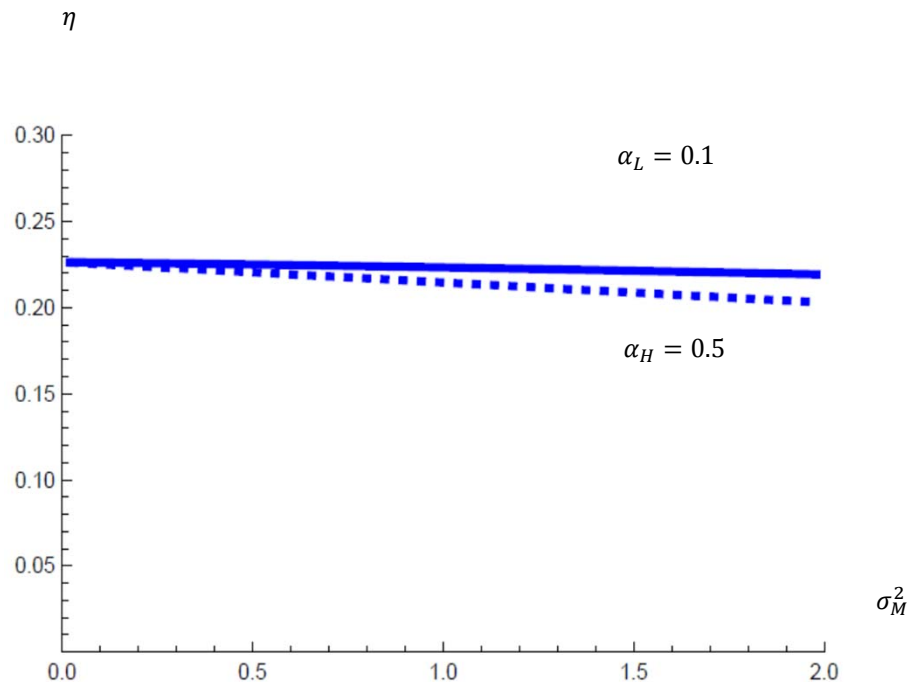


Theory

- ◆ Two channels through which an increase in political uncertainty leads to a reduction in households' stock market participation:
 - 1) an increase in asset risk reduces the speculative demand; and
 - 2) an increase in labor income risk increases the hedging demand to sell the stock when the stock return is correlated with labor income

Theory

- ◆ Heterogeneous effect: across different households, those with higher labor income exposure demonstrate a greater reduction in participation as a reaction to upcoming elections



- α is the labor income exposure parameter.
- σ_M^2 captures political uncertainty
- η is the participation rate.



Data



◆ Household data:

- Survey of Income and Program Participation (SIPP) data of the US Census Bureau

◆ Gubernatorial election data:

- Correlates of State Policy Project initiated by the Institute for Public Policy and Social Research (IPPSR)

◆ State unemployment:

- Bureau of Labor Statistics (BLS)

◆ State Housing price index:

- Federal Housing Finance Agency

◆ State GDP growth:

- Bureau of Economic Analysis (BEA)



SIPP

- ◆ Survey follows new sets of households up to four years
- ◆ Core set of questions (answered every four months):
 - demographics, employment and income, and business ownership
- ◆ Topical modules (answered annually):
 - assets and liabilities
 - ownership and market value of different types of assets, including real estate, vehicles, financial assets
 - Each rotation group is interviewed during months between May and November.
- ◆ Our sample includes 152,095 unique households with household heads' age +18
- ◆ 1996, 2001, 2004 and 2008 panels covering 1996-2000, 2001-2003, 2004-2007, and 2008-2013 waves



Summary Statistics: SIPP

Variables	No. of Obs.	Mean	Median	Std. dev.
% Households holding stocks	359,260	0.223	0.000	0.416
% Households holding stocks (inc. retirement accounts)	359,260	0.387	0.000	0.450
% Stock share (% of liquid wealth)	359,260	0.104	0.000	0.271

- ◆ Our analysis focus on stock investments outside of retirements accounts for several reasons
 - Households do not actively rebalance or trade in their retirement accounts (Agnew, Balduzzi, and Sunden, 2003; Mitchell et al., 2006; Benartzi and Thaler, 2007)
 - Withdrawals often incur significant penalties
 - Default choices affect investments in the retirement accounts (Beshears et al., 2009)



Gubernatorial elections data from IPPSR

◆ Elections:

- pre-scheduled and staggered across states and years held every four years, with the exception of Vermont and New Hampshire, which choose to run their gubernatorial elections every two years
- 36 states are subject to various term limits

◆ We exclude North Dakota, South Dakota, Maine, and Vermont because the SIPP mask these small states to protect confidentiality of respondents (merge Dakotas together and Maine and Vermont together)



Gubernatorial elections and SMP

Baseline model (Difference-in-differences)

$$StockMktPart_{i,s,t} = \beta_0 + \beta_1 Election_{s,t} + \mathbf{X}'_{i,s,t} \beta_2 + \delta_s + \mu_t + \alpha_i + \varepsilon_{i,s,t}$$

- ◆ Households in states without an upcoming election in a given year t is the control group for a treated sample of households in states about to elect a governor during the same year t
- ◆ Controls
 - household variables (financial wealth, total wealth, age, education level, marital status, financial occupation, race, and gender), and
 - state-level variables (GDP growth, unemployment rate, and housing price index (HPI))
 - Fixed effects (state, year, household)
- ◆ Standard errors are clustered at state and year levels to account for the correlations in households' decisions to participate in the stock market from the same state and for the correlations in the same year



Gubernatorial elections and SMP

Results of the baseline model

	Participation		% Stock share	
Election	−0.008** (−2.216)	−0.007* (−1.875)	−0.005*** (−2.794)	−0.006** (−1.980)
Presidential	−0.003* (−1.792)		−0.017*** (−2.928)	
Nobs	306,648	306,648	306,648	306,648
State fixed effects	yes	yes	yes	yes
Year fixed effects	no	yes	no	yes
Household fixed effects	yes	yes	yes	yes

- ◆ political uncertainty negatively and significantly affects households' stock market participation
- ◆ −0.008 and −0.007 imply a decrease of 3.1% and 3.5% in the average probability of stock market participation (22.3%)
- ◆ −0.005 and −0.006 imply a 4.8% to 5.8% decrease in the average % of liquid wealth invested in the stock market (10.4%)



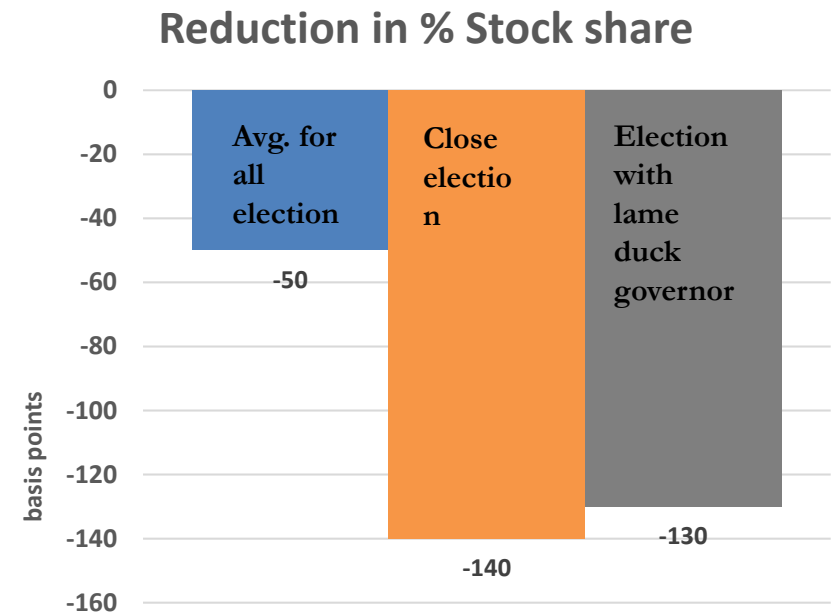
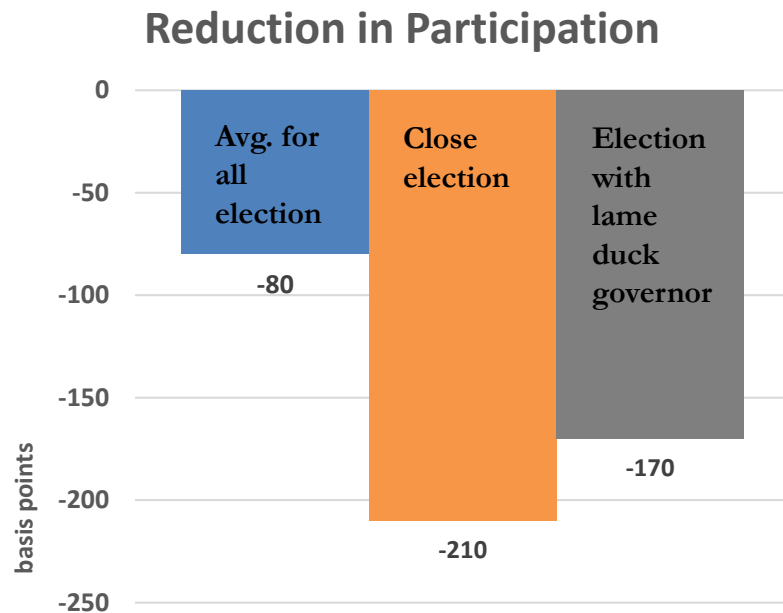
Elections associated with greater uncertainty

- ◆ Greater political uncertainty should have stronger effects on households' SMP
- ◆ Two proxies:
 - **Close elections:** victory margin in the lowest tercile (avg. 3.84% vs. 22.59% for non-close elections). Robustness: ex-ante measure of closeness constructed with pre-election poll data.
 - **Elections where incumbent governors cannot stand for reelections due to term limits:** Due to incumbency advantage, incumbents overwhelmingly win reelections. (83% of time in our sample). PU can exacerbate when the incumbent is in his/her last term. Term limits are also plausibly exogenous.



Elections associated with greater uncertainty: results

Estimated effects of PU on SMP



Effects are stronger for elections with greater uncertainty



How do households reallocate their assets?

	% Stock share ^W		% Safe share ^W		% Non-liquid ^W	
Election	-0.001**	-0.001**	0.012**	0.010*	-0.009	-0.012
	(-2.078)	(-2.369)	(2.247)	(1.809)	(-0.988)	(-1.035)
Nobs	306,648	306,648	306,648	306,648	306,648	306,648
State fixed effects	yes	yes	yes	yes	yes	yes
Year fixed effects	no	yes	no	yes	no	yes
Household fixed effects	yes	yes	yes	yes	yes	yes

- ◆ These three variables are scaled by total wealth to 1) control for any shocks to other parts of the households' portfolio, and 2) to avoid the mechanical relation that a decrease in the % of liquid wealth invested in stocks always indicates an increase in the % of liquid wealth invested in safe assets.
- ◆ Households dampen their stock investments by 4.4% (at the mean of 2.7%), while boosting their investments in relatively safer assets by 5.7% (at the mean of 19.1%).



Asset risk and labor income risk

- ◆ Households alter their SMP to reduce exposure to asset risk and to hedge the labor income risk when they face political uncertainty
- ◆ Two tests are employed to test this prediction:
 - exploits the cross-sectional variation in the in-state versus out-of-state stock investments of households that have elections in their states
 - Exploits the cross-sectional variation in the exposure of the households' to labor income risk



In-state vs. Out-of-state investments

- ◆ The effect of PU on “in-state” investments is expected to be stronger than their investments in other states (“out-of-state”) that do not have upcoming elections.
 - Gubernatorial elections are likely to increase the risk of in-state firms more than out-of-state firms (Bird, Karolyi, and Ruchti, 2017; Boone, Kim, and White, 2017) => Reduces the attractiveness of in-state stocks, and therefore reduces the speculative demand of households
 - Correlation between households’ labor income and the performance of in-state firms is higher than the correlation between households’ labor income and the performance of out-of-state firms => To hedge the exposure of labor income to political uncertainty, households have greater incentives to reduce their investments in in-state firms than in out-of-state firms



In-state vs. Out-of-state investments

- ◆ Data on households' equity holdings from a large discount brokerage firm for the period 1991 to 1996
- ◆ These data provide monthly information on common stock holdings for a large panel of households residing in different states
- ◆ Examine the investment behavior of households in their in-state and out-of-state stocks around elections



In-state vs. Out-of-state investments

	(1) In-state investment	(2) Out-of-state investment
Election	-0.077** (-2.267)	-0.018 (-0.715)
State GDP growth	0.032*** (5.356)	0.004 (1.004)
State unemployment	0.026** (2.009)	0.008 (1.094)
State HPI appreciation	0.002 (0.260)	0.003 (0.736)
Year fixed effects	Yes	Yes

Consistent with the theoretical prediction, we find that gubernatorial elections have a significantly greater negative impact on in-state investments than on out-of-state investments



Households with different labor income exposures to PU

- ◆ Households with a greater labor income exposure to PU have stronger incentives to hedge the exposure => the reduction in SMP to be more pronounced for these types of households
- ◆ State and local government employees
 - Cahan (2017) finds that governors and their party allies may have the ability to raise employment levels leading up to elections, or delay employment-reducing decisions until afterwards
 - In the two quarters following the election, state and local government employment growth is generally lower by similar magnitudes in those counties that have just experienced an election compared to those that did not, and such effects are absent in private sector employment
- ◆ Self-employed in politically sensitive industries, including transportation, warehousing, utilities, public administration, educational, health and social services, and mining, are more susceptible to changes in political landscape than others (Kostovetsky, 2015)



Households with different labor income exposures to PU

Model

$$StockMktPart_{i,s,t} = \phi_0 + \phi_1 Election_{s,t} \times \mathbf{Z}_{i,s,t} + \mathbf{Z}'_{i,s,t} \phi_2 + v_{s,t} + \alpha_i + \varepsilon_{i,s,t}$$

- ◆ Coefficients of interest are the interaction terms
- ◆ Include state×year fixed effects to control for any time-varying state conditions
- ◆ Identify variations in stock market participation across households residing in the same state at the same point in time
- ◆ Election itself is absorbed by the statexyear fixed effect



Households with different labor income exposures to PU

	Participation	% Stock share
<i>Labor income risk</i>		
Government employee × Election	−0.004*** (−2.883)	−0.003*** (−3.093)
Business owner in PSI × Election	−0.003* (−1.810)	0.002 (0.903)
Nobs	306,648	306,648
Household fixed effects	Yes	yes
State-year fixed effects	Yes	yes

These results support the prediction that households with higher labor income risk (those employed in public sectors and business owners operating in politically sensitive industries) are particularly sensitive to changes in political uncertainty



Households with different labor income exposures to PU (contd.)

	Participation	% Stock share
<i>Other demographic characteristics</i>		
College or more × Election	0.004*** (3.018)	0.003*** (3.100)
Some college × Election	0.003** (2.119)	0.001* (1.882)
Young × Election	0.003* (1.905)	0.002** (2.186)
Middle aged × Election	0.005 (1.536)	0.006* (1.699)
Total wealth × Election	0.002*** (4.527)	0.003*** (4.020)

- Well-educated and young households have more job mobility
- Wealthy households are less concerned about their labor income
Younger, wealthier, and better-educated households do not need to hedge the exposure of labor income to political uncertainty as much as their counterparts, resulting in a lower sensitivity of SMP to elevated political uncertainty



Dynamics of SMP during an election cycle

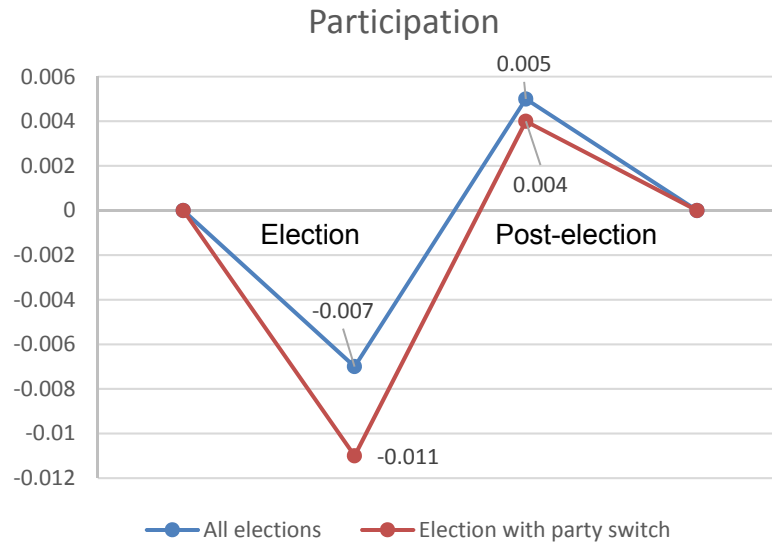
- ◆ If uncertainty is resolved after the election outcome, we expect the decline in participation to be temporary
- ◆ A complete reversal would suggest that there is only an intertemporal substitution of participation when households face uncertainty
- ◆ A partial reversal would indicate that uncertainty has a long-lasting and disruptive effect on participation



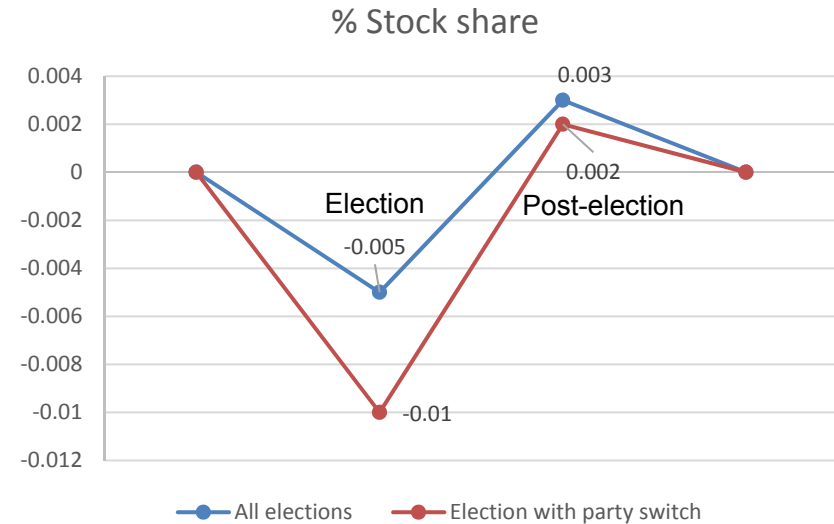
Dynamics of SMP during an election cycle (contd.)

- ◆ Magnitude of reversal should depend on the degree of resolution in political uncertainty after the election
- ◆ For elections where a new governor from a different political party is elected, we expect the political uncertainty to remain comparatively high
 - Different parties are likely to have different political ideologies and classes of constituents, which can lead to differences in their stances on policy positions and political actions (Hibbs, 1977; Alesina, 1987; Alesina and Sachs, 1988).

Dynamics of SMP during an election cycle: results



Participation: All elections	Participation: Party switch
-0.002	-0.007***



%Stock share: All elections	%Stock share: party switch
-0.002	-0.008***



Conclusions and implications

- ◆ Provide new evidence on the effect of political uncertainty on households' decision to participation in the stock market
 - Our theoretical framework reveals two channels through which PU reduces households' participation: asset risk (speculative demand) and labor income risk (hedging demand)
 - An increase in PU is associated with a significant decline in both the propensity and intensity of households to invest in the stock market
 - Households reallocate their capital to safer assets
 - Households' labor income exposure to political uncertainty help explain the differential sensitivities of households' SMP to political uncertainty
 - Reversal in households' SMP after the election - magnitude of reversal depends on the level of uncertainty after the election
 - In case of a party change, reversal is less than complete implying a long-lasting and disruptive effect of uncertainty on participation



Conclusions and implications (contd.)

- ◆ Important implications for households, firms, and economy in general:
 - Since the equity risk premium is positive, lack of participation in the stock market can have significant negative effects on households' wealth accumulation and retirement savings
 - If the demand for stocks is lower during periods of high uncertainty, it can raise the costs of raising capital, which might delay firms' issuance of equity as well as corporate investments -- can worsen or slow down recovery from economic recessions as periods of high political uncertainty and economic downturns tend to coincide
 - Our finding that wealthier households tend to reduce their SMP less after periods of high political uncertainty suggests that such households can benefit from the equity premium in the long run, in contrast to the poor and middle class households